MT-116 (Calculus & Analytical Geometry)

Assignment (10 marks)

[CLO-02] **[03]**

Q. 1 A rescue drone is flying over a disaster zone to drop medical supplies. Its motion is controlled, so that it follows a smooth curved path to avoid obstacles. The path of the drone is modeled by:

$$x = t^2 - 1, \qquad y = t^3 - 3t$$

At a critical moment $t = 2$, the engineers need to check whether the turn is too sharp for save navigation? To measure this, they calculate the radius of curvature. Find the radius of curvature at $t = 2$.

[CLO-02] **[03]**

Q. 2 A company is designing a signal tower system for a drone network. The signal strength depends on two design parameters' x and y, given by:

$$S(x, y) = xy$$

However, due to energy limits, the system must satisfy the constraints $x^2 + y^2 = 25$.
Find the values of x and y that maximize the signal strength.

[CLO-03] **[04]**

Q.3 (a) Model a real-life situation as a sequence or series.

(b) Identify which convergence test (included in the course) is appropriate and apply the test.

(c) Conclude the convergence behavior and interpret the results in real-life.